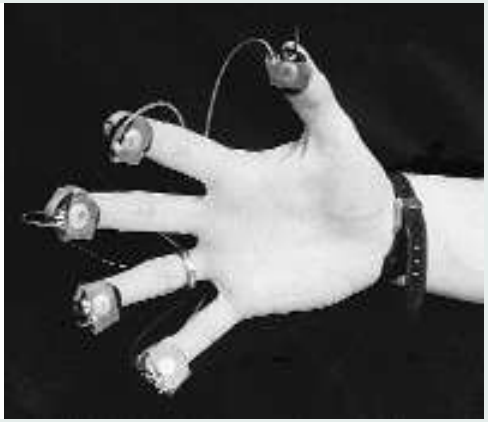
Embedded Control for Robotics



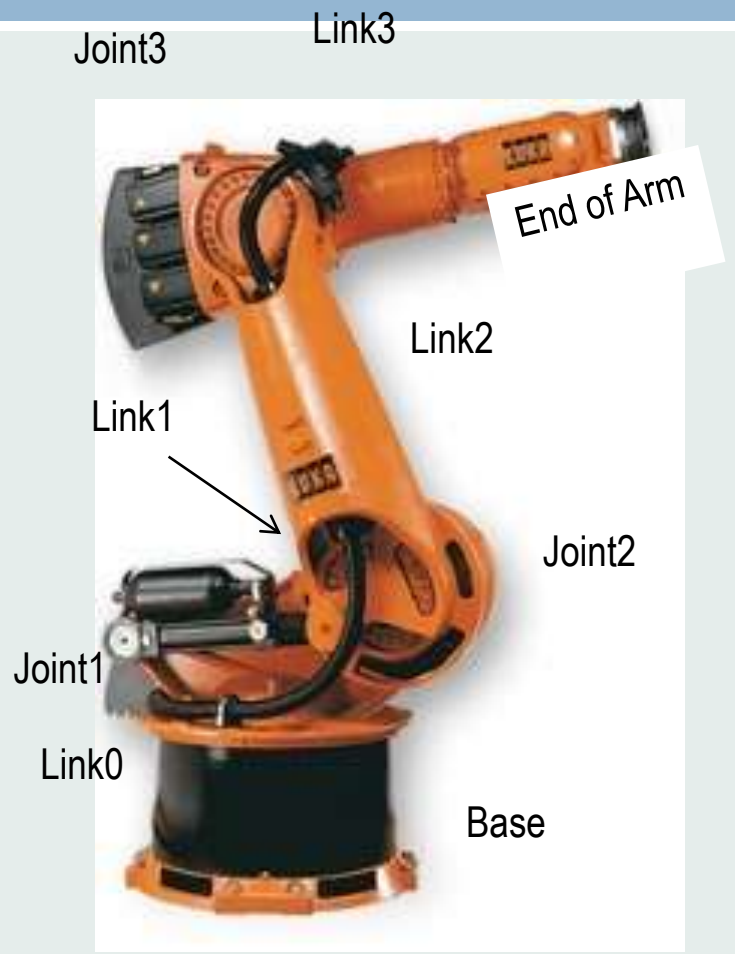
Dr. Suleman Khan
CS, NUCES-FAST, Islamabad

Outline

- Introduction
- Motivation
- Robots Components
- Course Learning Outcome
- Background Terminology

What is a Robot ?

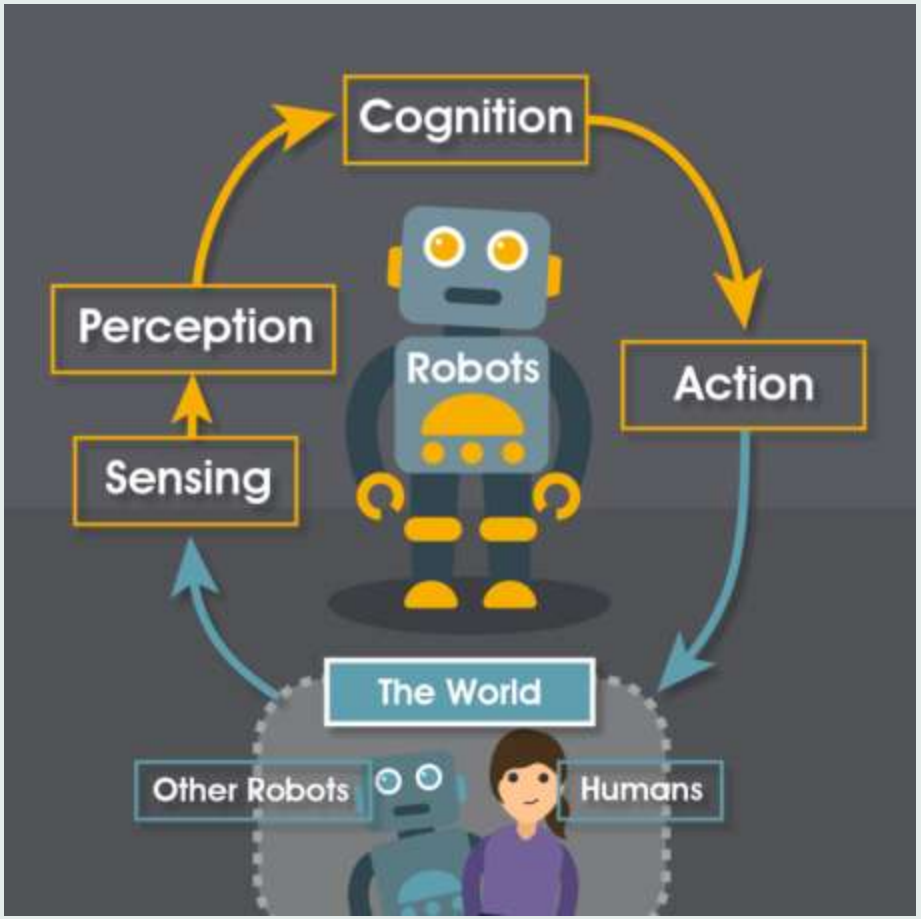
- A robot is a re-programmable, multifunctional manipulator designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.(*Robotics Association of America*)
- A machine that resembles a human being and does mechanical routine tasks on command(*Random House Dictionary*).



What is a Modern Robot ?

An electromechanical device that is:

- Reprogrammable
- Multifunctional
- Sensible for environment or
- perceive the environment
- reason about events
- make plans using algorithms
- perform actions by actuators



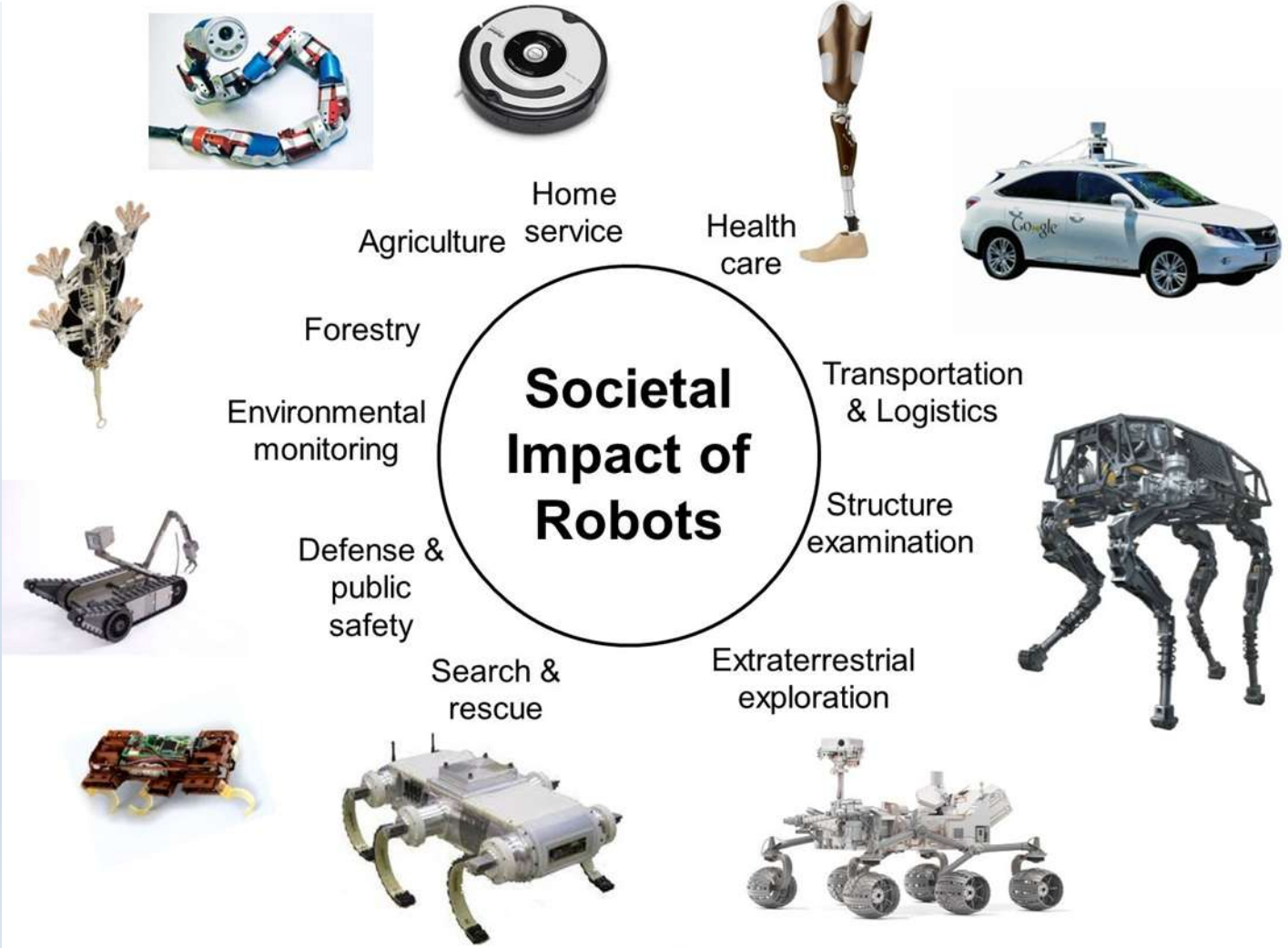
What is Robotics ?

- Robotics is the art, knowledge base, and the know-how of designing, applying, and using robots in human endeavors.
- Robotics is an interdisciplinary subject of
  - **Mechanical Engineering** – concerned primarily with manipulator/mobile robot design, kinematics, dynamics, compliance and actuation.
  - **Electrical Engineering** – concerned primarily with robot actuation, electronic interfacing to computers and sensors, and control algorithms.
  - **Computer Science** – concerned primarily with robot programming, planning, and intelligent behavior(AI),etc...

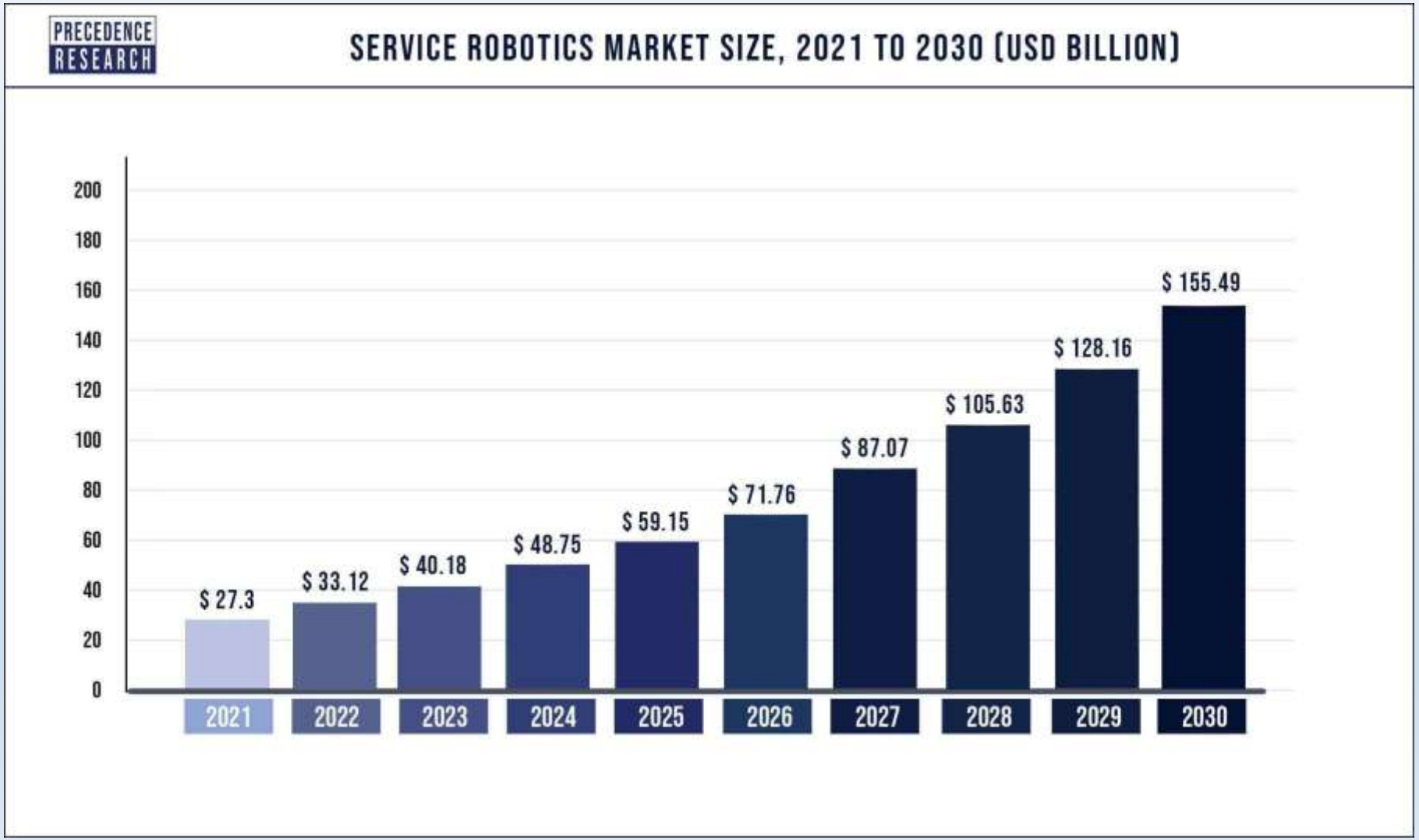
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Robots are everywhere

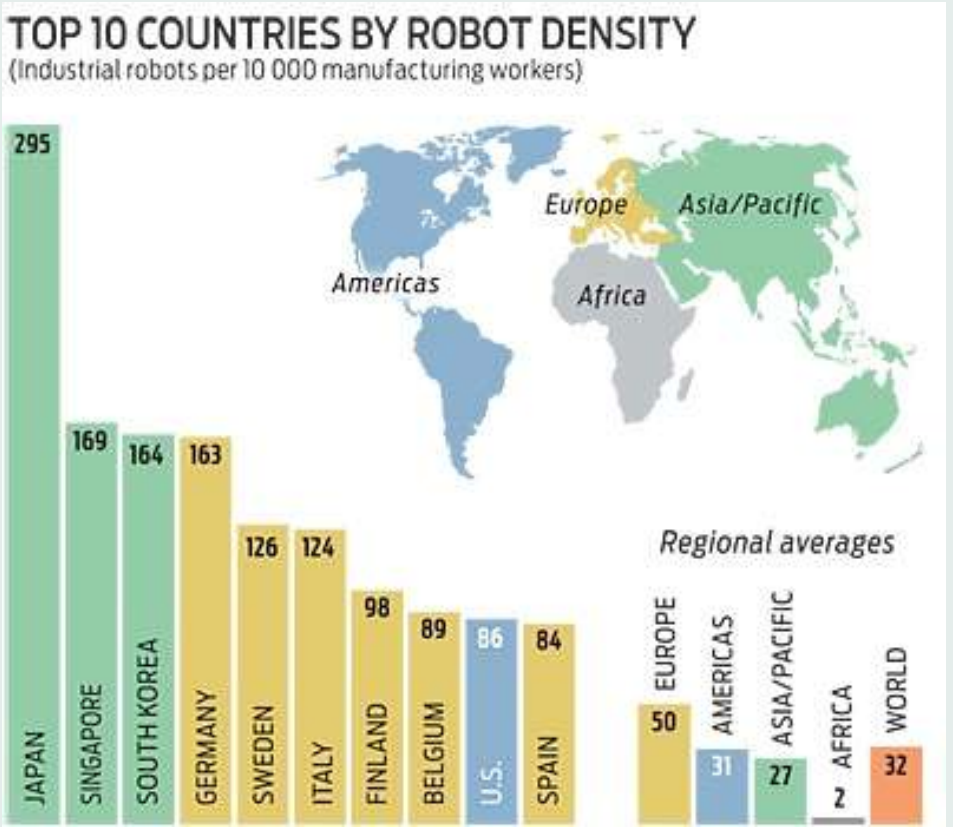


World market of Robotics!

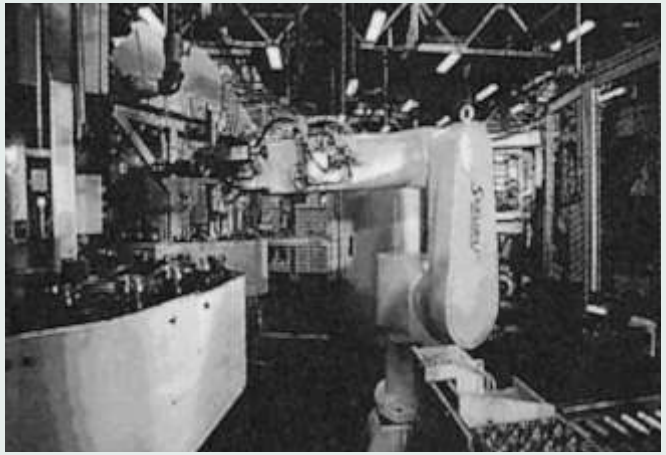


Robot applications

1. Material handling applications
   - Material transfer – pick-and-place, palletizing
   - Machine loading and/or unloading
2. Processing operations
   - Welding
   - Spray coating
   - Cutting and grinding
3. Assembly and inspection
   - Cutting and grinding
   - Military
4. Medical Application
   - Surgery
   - Tele-Operation



Robots Material handling



A robot loading and unloading



A robot placing dishwasher tubs



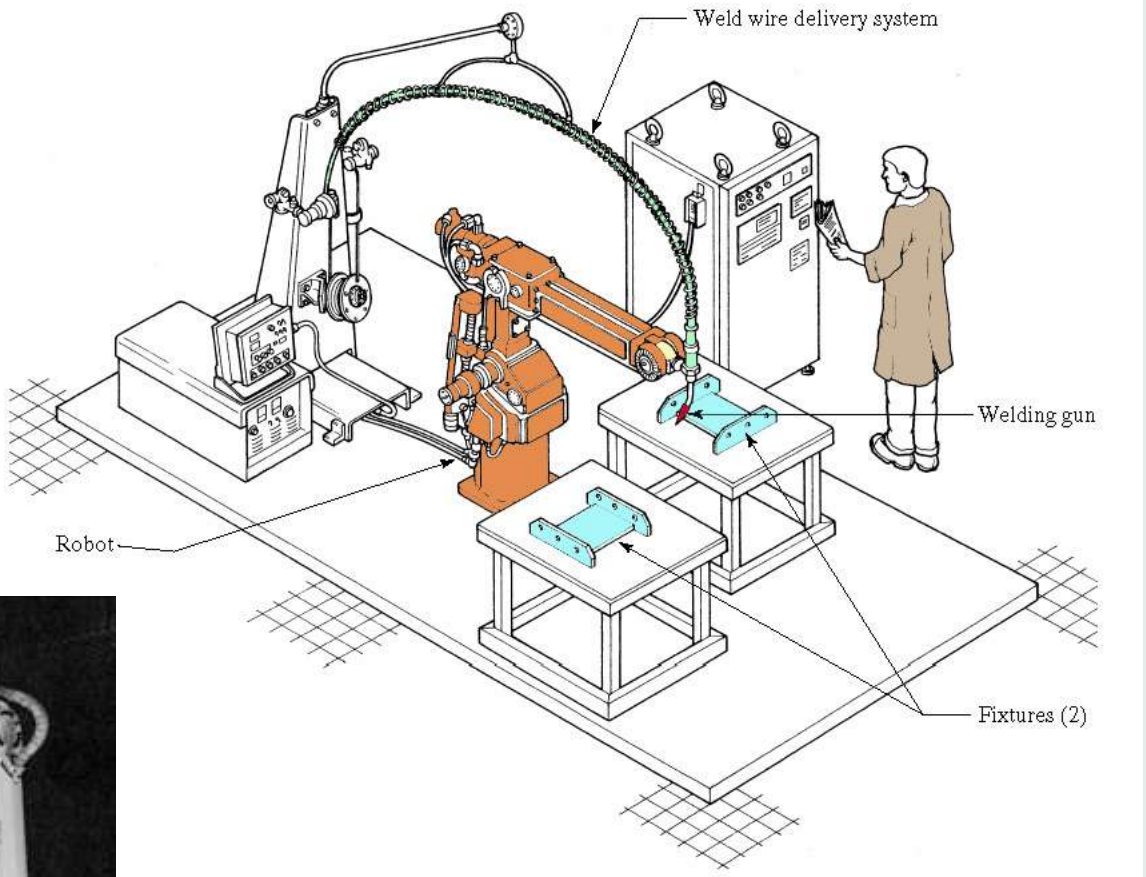
A P200 Fanuc painting automobile bodies



Space Exploration

Robotic Arc welding cell

- Robot performs flux-cored arc welding (FCAW) operation at one workstation while fitter changes parts at the other workstation





Robots in Military

- Computer GPS guided bombs "smart bombs".
- Movement of troops, weapons, etc.
- Advancements in war weapons, aircraft, artillery, naval ships, etc.



Robots in medical



Medical Robot of German



Da Vinci Robotic Surgery System

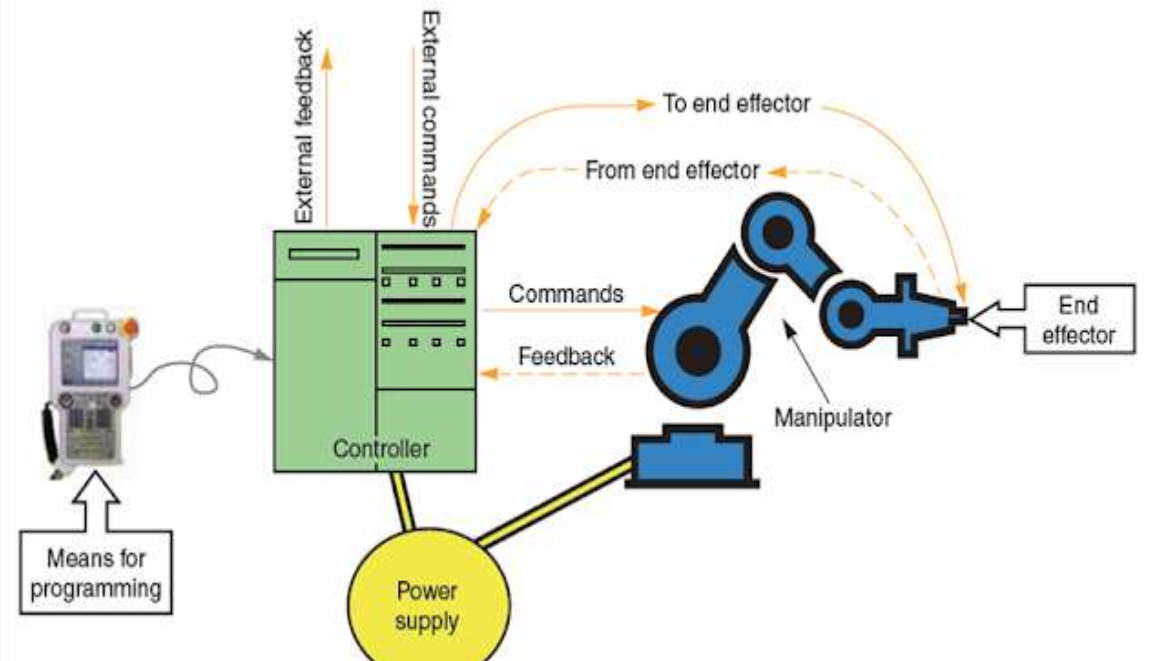
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Robot Components

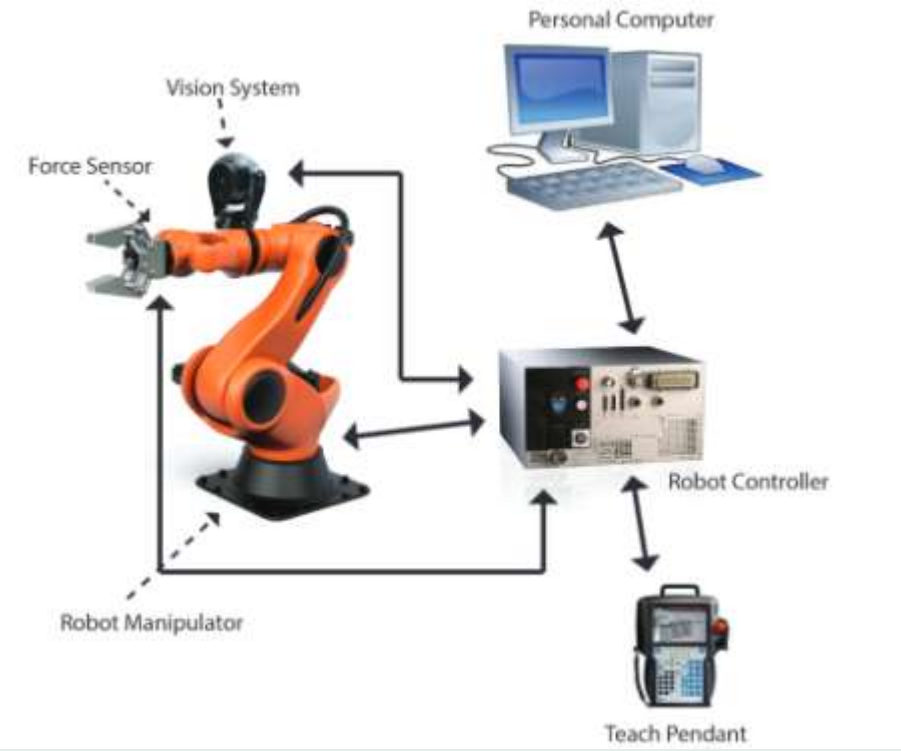
- Manipulator
  - Actuators/Links
  - End Effectors
  - Sensors
- Controller
- Power Source
- Software



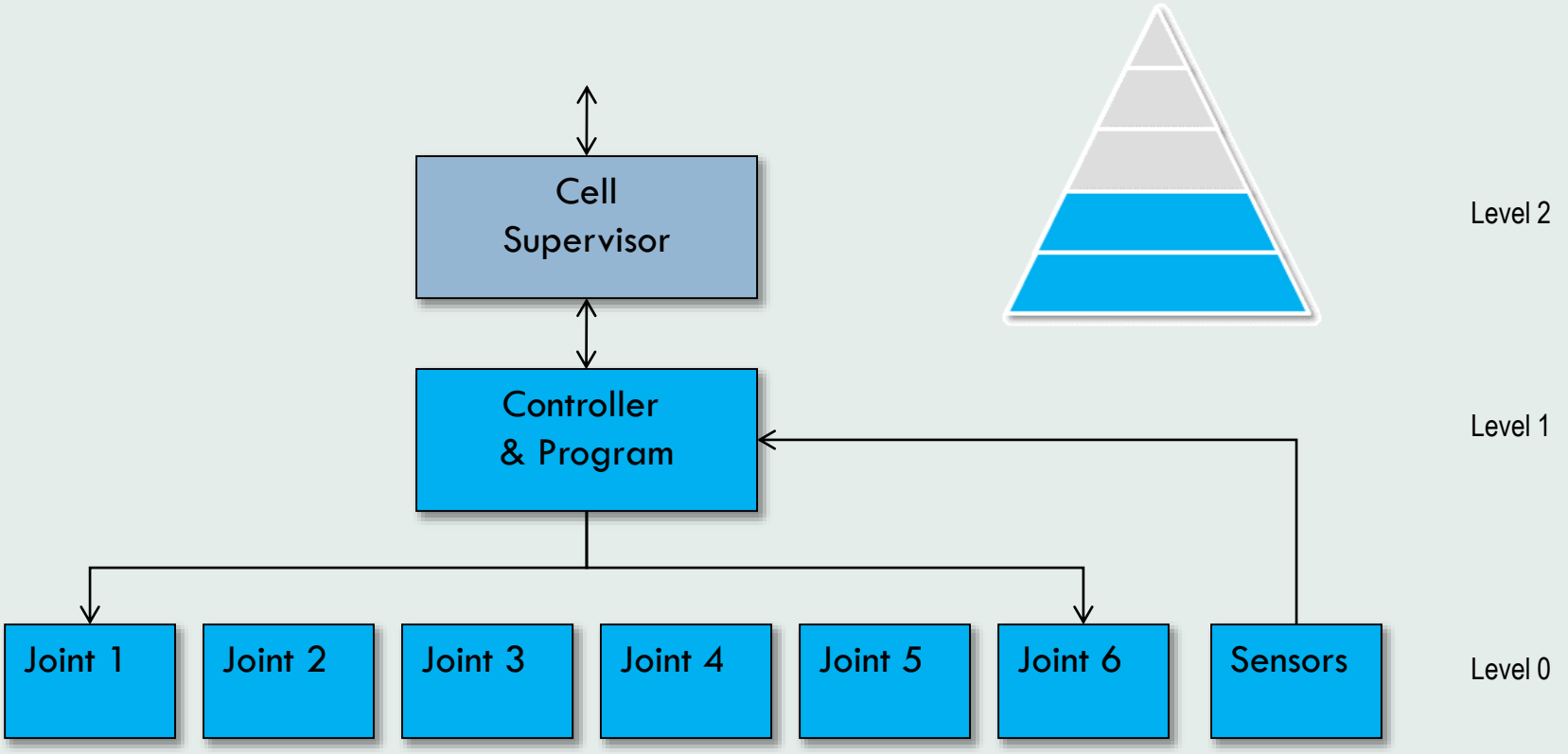


Robot Control Systems

- Issues instructions to the robot to perform the desired tasks.
- Controls peripheral devices.
- Interfaces with robot.
- Interfaces with humans.



Robot Control Systems

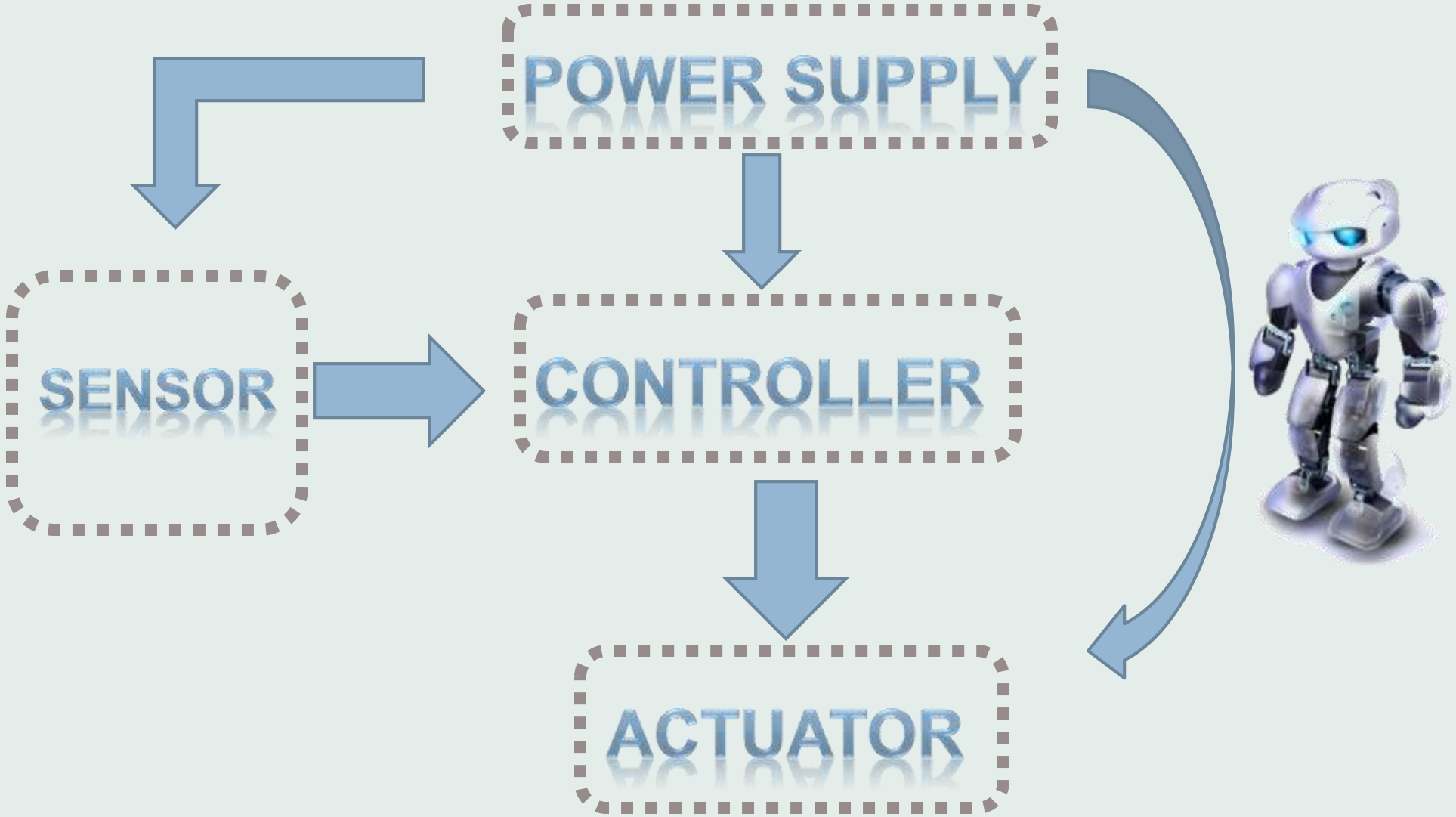


Robot Control Systems

- **Limited sequence control** – pick-and-place operations using mechanical stops to set positions
- **Playback with point-to-point control** – records work cycle as a sequence of points, then plays back the sequence during program execution
- **Playback with continuous path control** – greater memory capacity and/or interpolation capability to execute paths (in addition to points)
- **Intelligent control** – exhibits behavior that makes it seem intelligent, e.g., responds to sensor inputs, makes decisions, communicates with humans

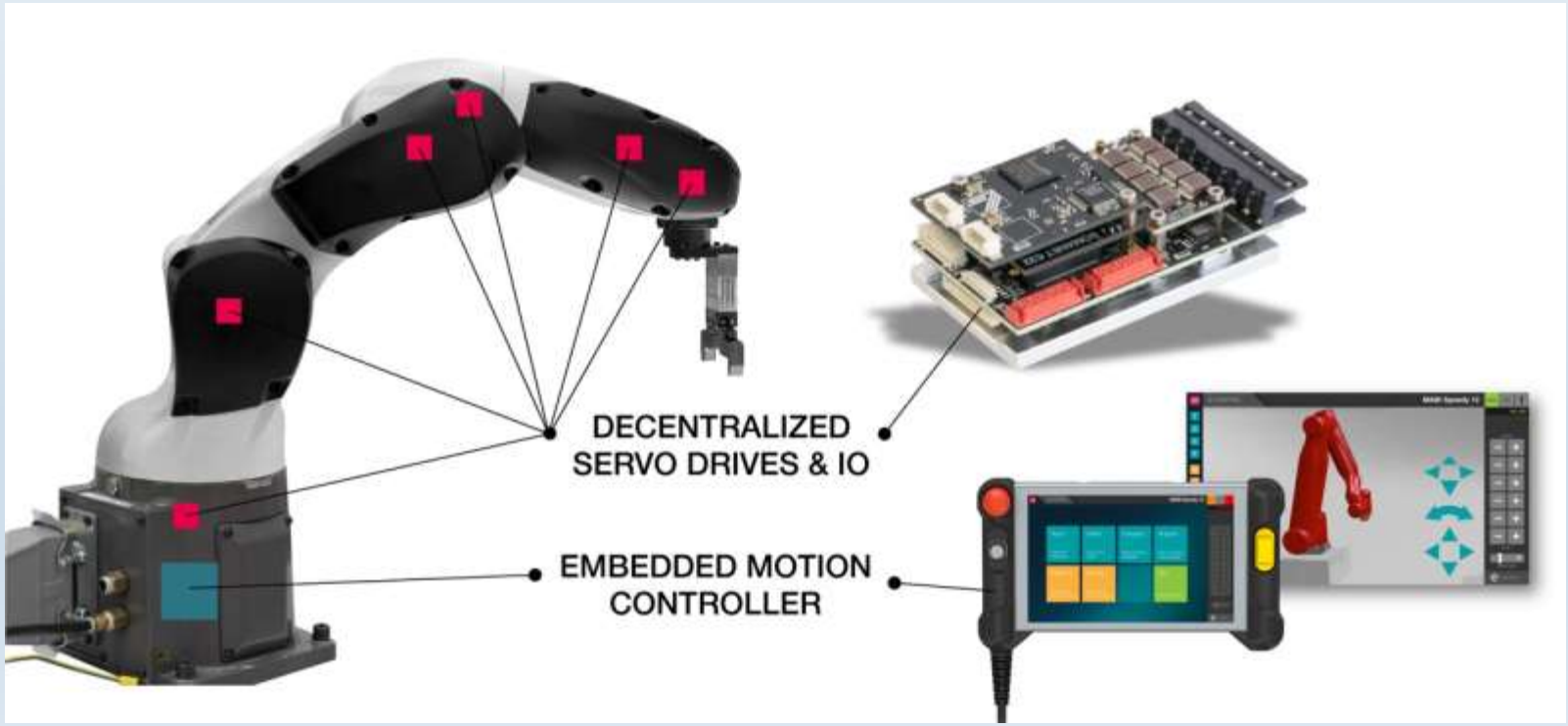


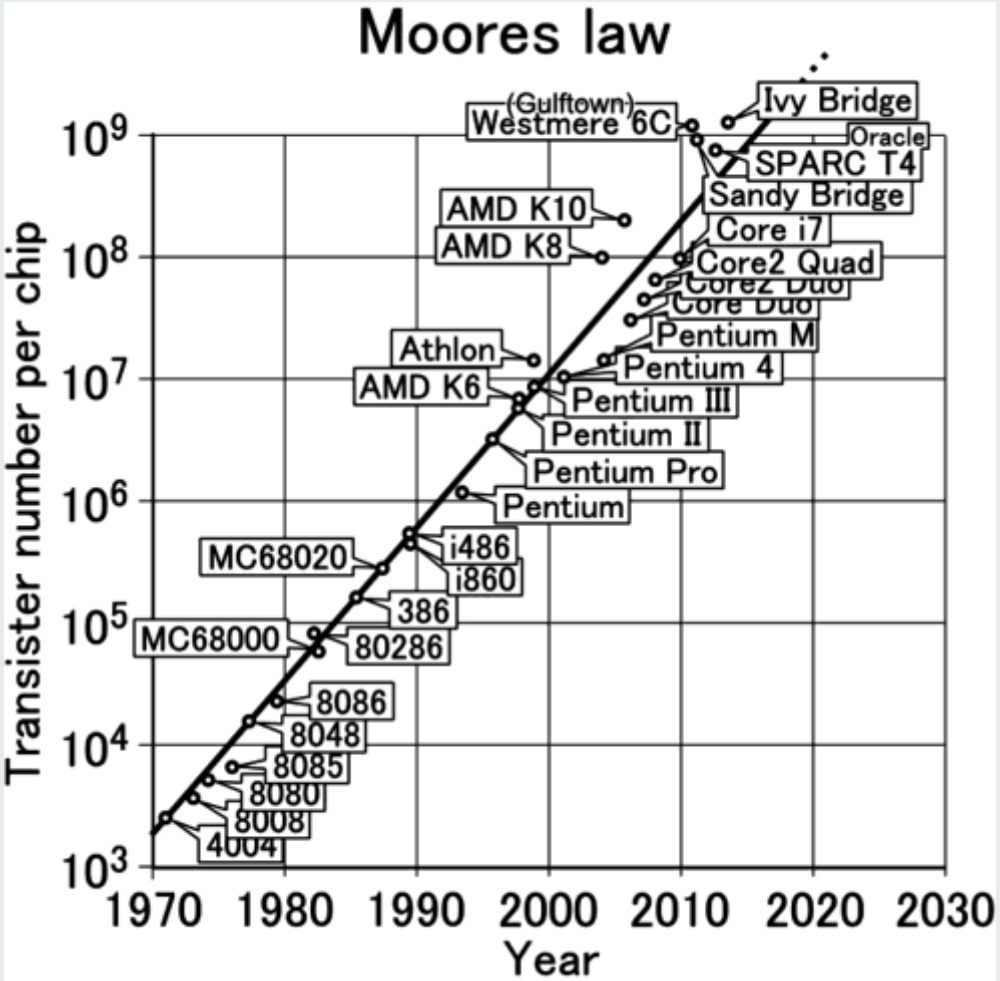
Robot Components working



Embedded Controllers for Robots

- An Embedded Controller (EC) is a microcontroller in computers that handles various control tasks.
- This is possible due to technology trends




Moores law
Transister number per chip
Year
10^3
10^4
10^5
10^6
10^7
10^8
10^9
1970
1980
1990
2000
2010
2020
2030
4004
8008
8080
8085
8048
8086
80286
386
i860
i486
Pentium
Pentium Pro
Pentium II
Pentium III
Pentium 4
Pentium M
Core Duo
Core2 Duo
Core2 Quad
Core i7
Sandy Bridge
SPARC T4
Oracle
Ivy Bridge
(Gulftown)
Westmere 6C
AMD K10
AMD K8
Athlon
AMD K6
MC68020
MC68000

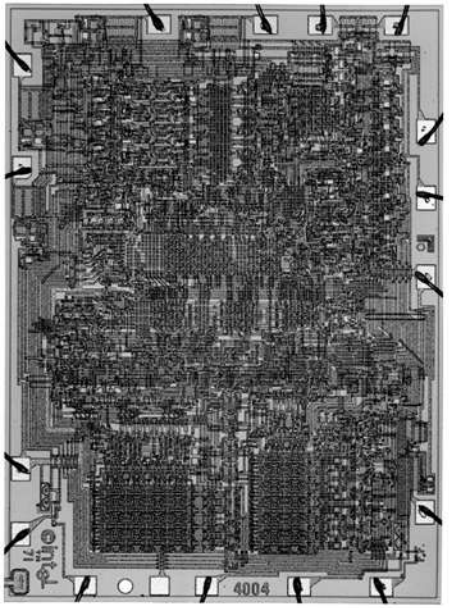
25 years of technology advance

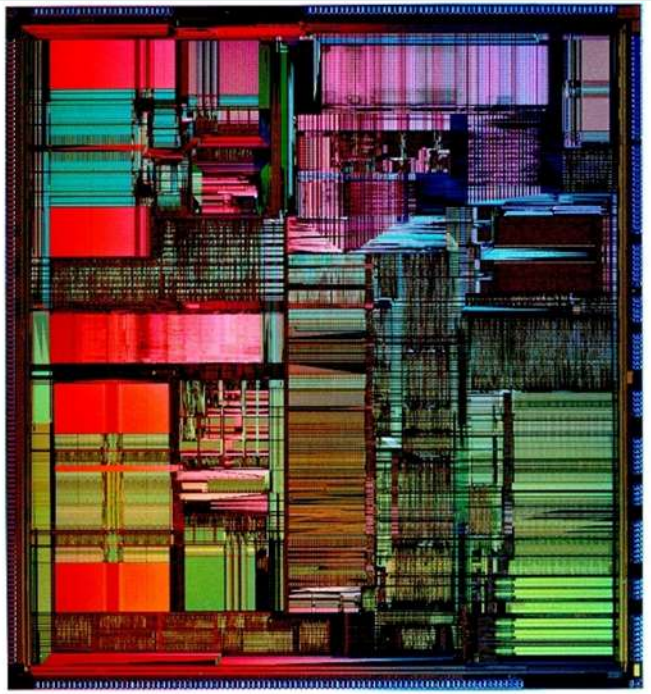
INTEL 4004 (1971)

4-bit data

2300 transistors, 10 microns

0,06 MOPS, 108 kHz





INTEL Pentium II (1996)

32-bit data

5.5M transistors, 0.35µ, 2 cm^2

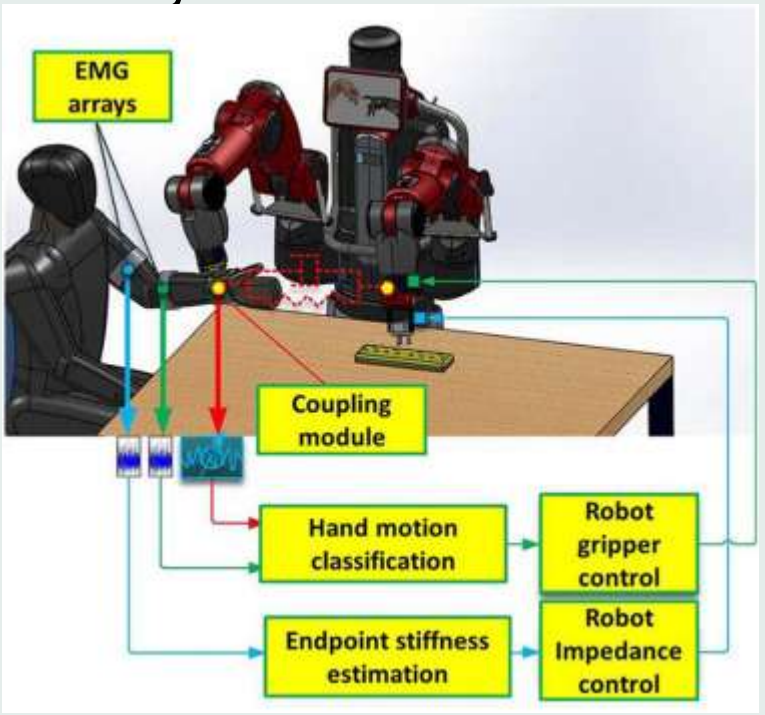
200 MHz, 200 MOPS, 3.3V, 35W

Trends

- In the past, mobile robots were controlled by large and expensive computer systems

- Today, we can build small mobile robots with numerous actuators and sensors that are controlled by
  - inexpensive,
  - small
  - lightweight

embedded computer systems



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Course Learning Outcomes

1. **Describe** basic control system theory and applications.
2. **Analyze** control systems for stability and performance.
3. **Design** controllers for robotic systems for stability and performance enhancement.

Class communication

- Flex
- Google classroom
- Office: C-205-E
- Will be announced on GCR

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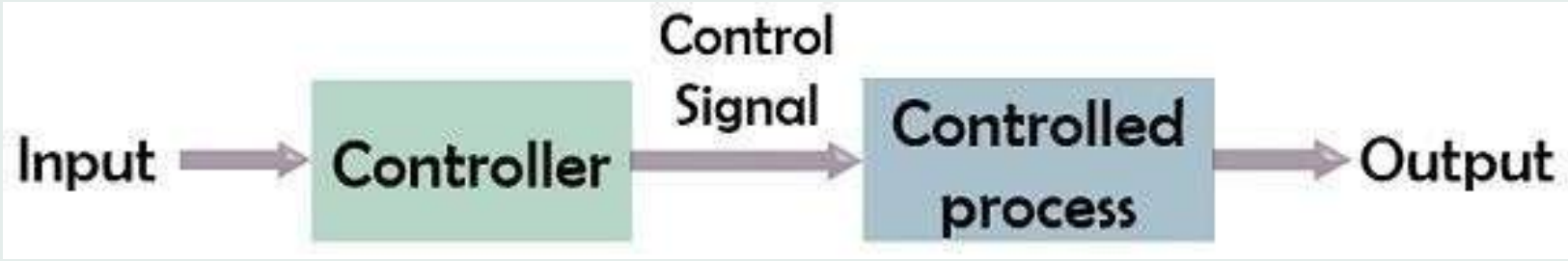
What is?

- System?
- Analog/Digital system?
- Control System?
- Microcontroller/Microprocessor?
- Embedded controller?
- Feedback in control
- Open Loop and Close Loop system?

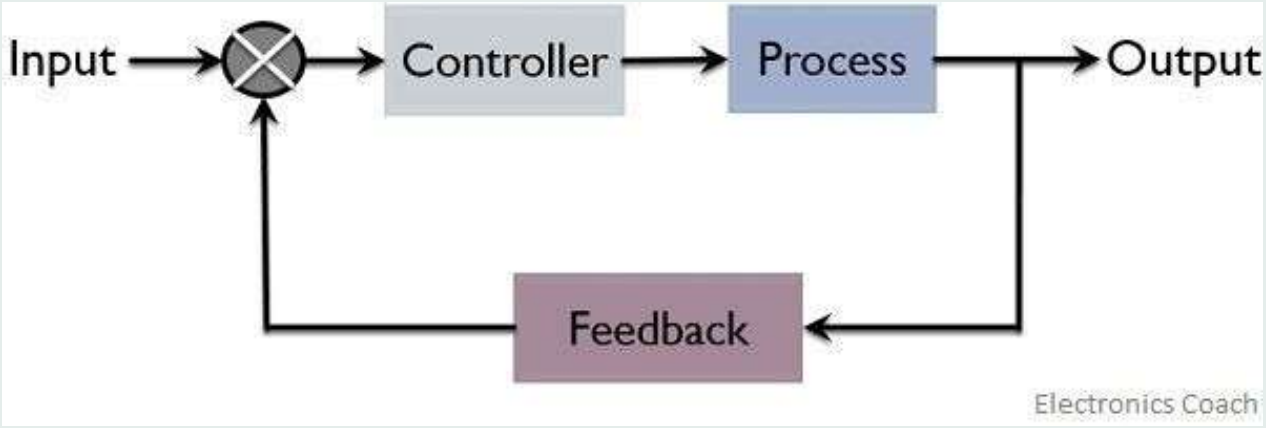
System and Control System

- System: A system can be defined as the combination of parts to carry out a specific task.
- Control is the act of producing a desired output for a given input. The control system is used for this purpose.
- Example: Controller is the logic that enables a drone to follow a desired trajectory.

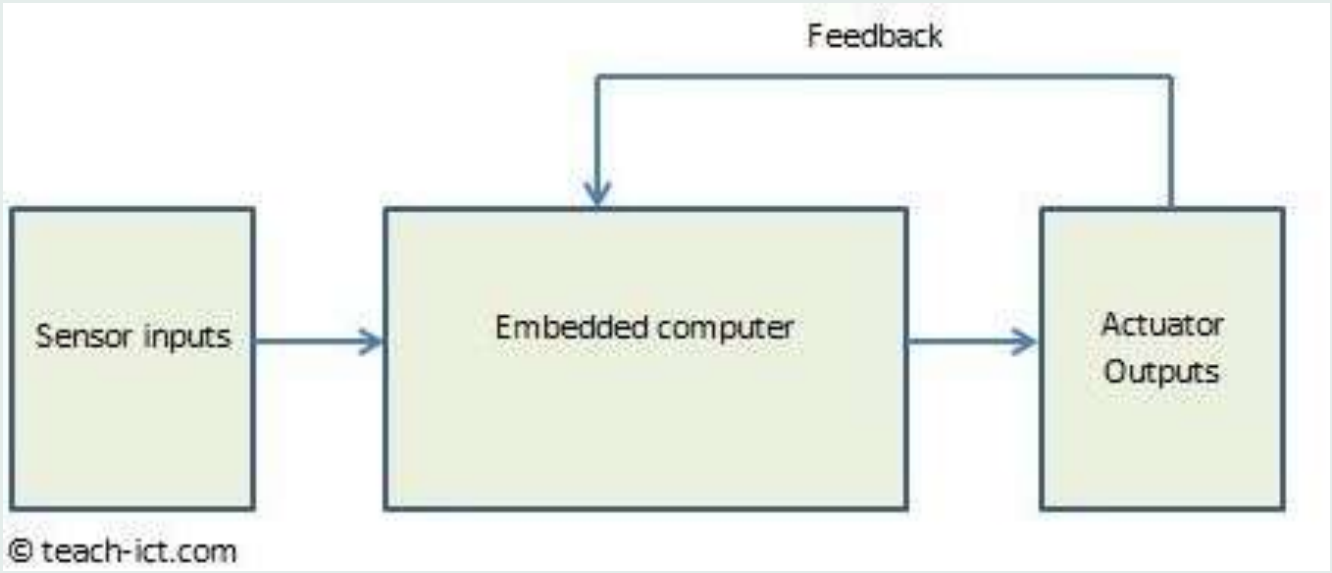
Open Loop System



Closed Loop System

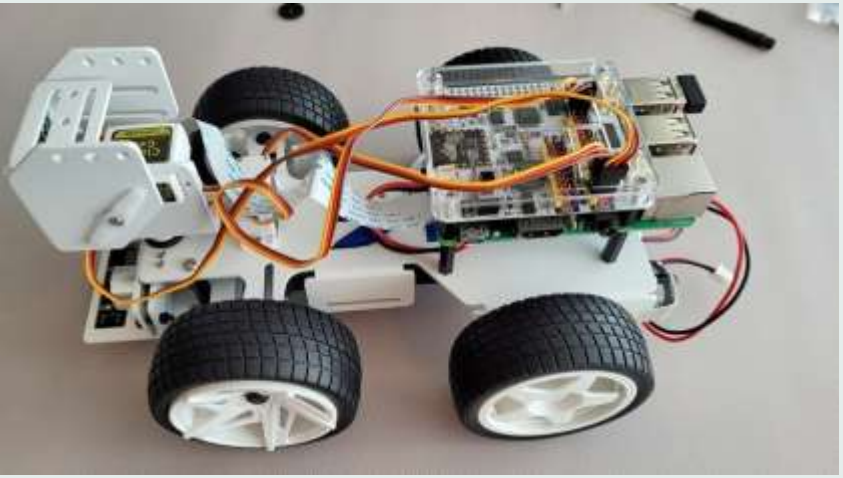


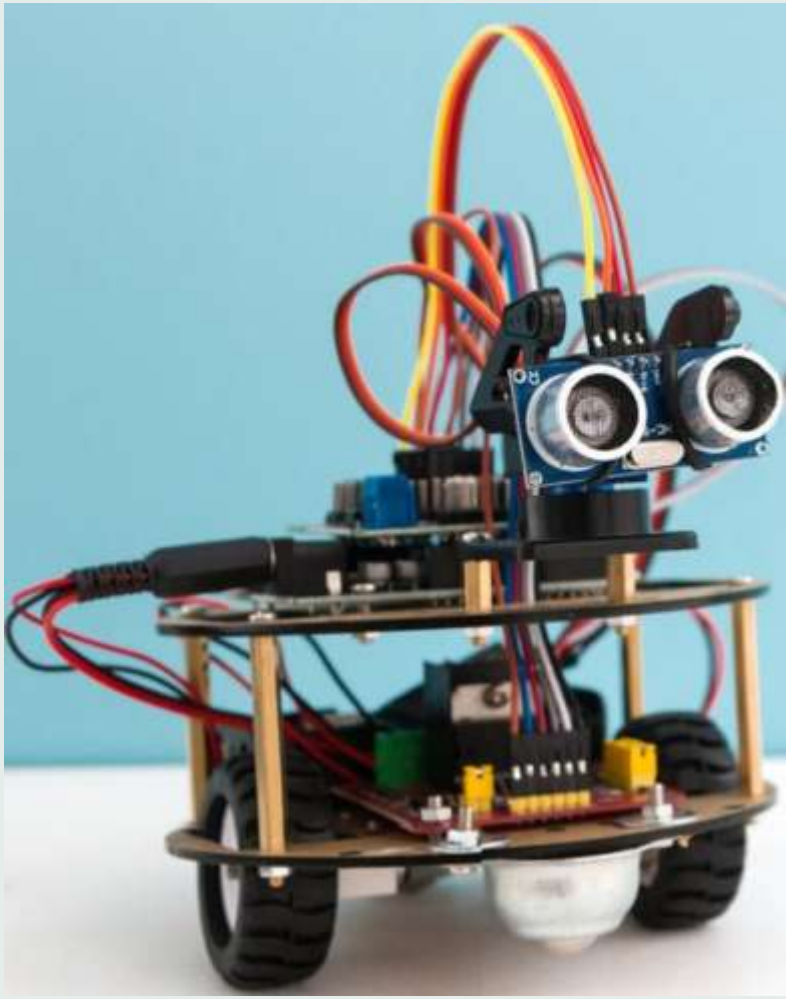
Feedback


Feedback
Sensor inputs
Embedded computer
Actuator
Outputs
© teach-ict.com

Questions

Project Kit





What Can Robots Do?

- **Increased productivity:** Repetitive jobs that are boring, stressful, or labor-intensive for humans(Welding Robot)
- **Safety:** Jobs that are dangerous for humans for example Decontaminating Robot- cleaning the main circulating pump housing in the nuclear power plant
- Repetitive work cycle with **Consistency and accuracy**





Continue…

- Manual tasks that human don’t want to do- **Dirty task**
- Process **multiple tasks** simultaneously.
- **Impossible things become possible:**-Assisting the handicapped
- **Cost savings:** By automating repetitive tasks, robots can reduce labor costs.

• Dirty tasks



• Assisting the handicapped



Cons & Future concerns

- **Loss of jobs** due to replacement by robots.
- **War** of the Worlds?



